MANYARA REGION

FORM FOUR MOCK EXAMINATION 2026

BASIC MATHEMATICS MARKING GUIDE

CODE: 041

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SECTION A (60 MARKS)

1. (a). Let a= $1.25^{\bullet}$---------(i)

10a=$12.5^{\bullet}$--------(ii)----------(01)

100a= $125.5^{\bullet}$-------(iii)

Equation (iii)--- equation (ii)

90a= 113-------------------------(01)

A=$\frac{113}{90}$

$1.25^{\bullet}$= $\frac{113}{90}$ ----------01

(b). 10% on renf, 20% 0n refreshment

25% on food, 15% on school fees

He was left with Tshs.150,000/=

10 + 25 + 20 + 15 = 70%

100 – 70= 30% ----------01mark

30% = 150,000/=
100%= X

30X= 150000 X 100 ----------01mark

X= $\frac{150000\ X\ 100}{30}$

X= 500,000

Johns salary= 500,000/= ---------- 01mark

2. (a). $\frac{\sqrt{63}+\sqrt{72}}{\sqrt{32}+\sqrt{28}} = \frac{\sqrt{9\,x\,7}+\sqrt{36\,x\,2}}{\sqrt{16\,x\,2}+\sqrt{4\,x\,7}}$ ----------0.5marks

$= \frac{3\sqrt{7}+6\sqrt{2}}{4\sqrt{2}+2\sqrt{7}}$ ----------0.5marks

Rationalize the denominator

$=(\frac{3\sqrt{7}+6\sqrt{2}}{4\sqrt{2}+2\sqrt{7}})\ (\frac{4\sqrt{2}-2\sqrt{7}}{4\sqrt{2}-2\sqrt{7}})$ ----------0.5mark

$= \frac{12\sqrt{14}-6\sqrt{49}+24\sqrt{4}-12\sqrt{14}}{16\sqrt{4}-8\sqrt{14}+8\sqrt{14}-4\sqrt{49}}$ ---------- 0.5mark

$= \frac{-42+48}{32-28}$ ----------0.5mark

$=\frac{6}{4}$

$= {}^{3}/_{2}$

$\frac{\sqrt{63}+\sqrt{72}}{\sqrt{32}+\sqrt{28}} = {}^{3}/_{2}$ ---------- 0.5mark

(b). $\log_2 x + \log_X 64 = 5$

$\log_2 x \log_X 2^6 = 5$

$\log_2 x + 6\log_x 2 = 5$

But $\log_b a = \frac{\log a}{\log b}$ ----------0.5 mark

$\log_2 x + \frac{6\log 2}{\log x} = 5$

$\log_2 x + 6\,{}^{1}/_{\log_2 x} = 5$

Let $\log_2 x$= M---------- 0.5mark

$M + {}^{6}/_{M} = 5$

$M^2 + 6 = 5M$---------- 0.5mark

$M^2 - 5M + 6 = 0$

$M^2 - 2M - 3M + 6 = 0$

$M(M-2) - 3(M-2) = 0$----------0.5mark

(M - 3) (M - 2)= 0

M= 2 or 3

$\log_2 x = 2$ ----------0.5mark

X=2^2

X= 4

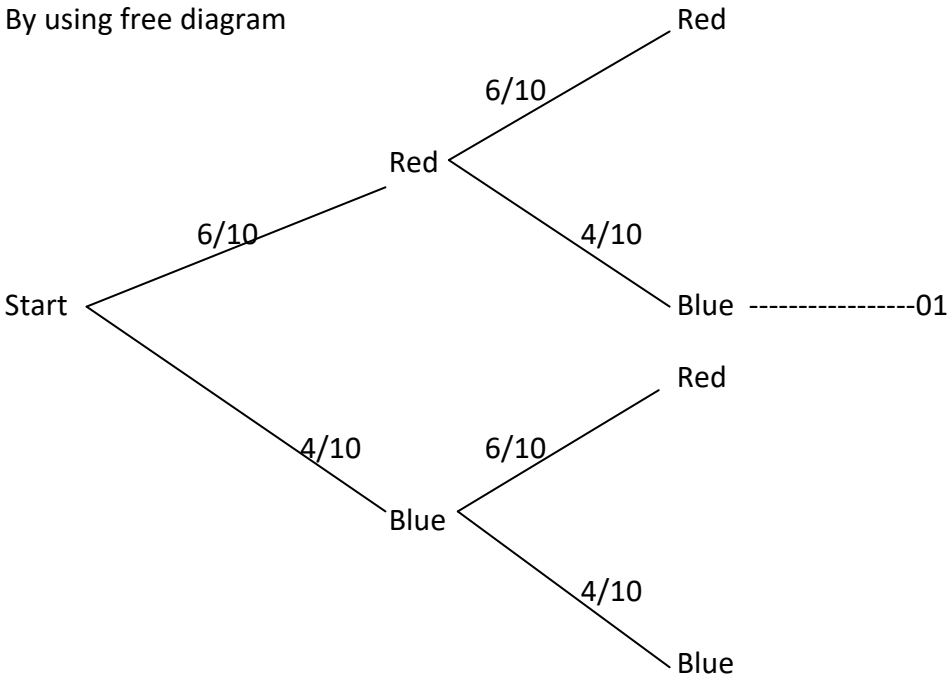
$\log_2 x$= 3----------0.5mark

2^3= X

X= 4 or 8 ---------- 0.5mark

3. (a). Samples spaces 6Red + 4blue = 10 marbles

By using free diagram



(i). Both of the same color

RR or BB

$^6/_{10}\ X\ ^6/_{10} +\ ^4/_{10}\ X\ ^4/_{10}$

$\frac{24}{100} + \frac{24}{100}$

$^{60}/_{100} = {}^{3}/_{3}$ ---------- 01mark

(ii). RB or BR

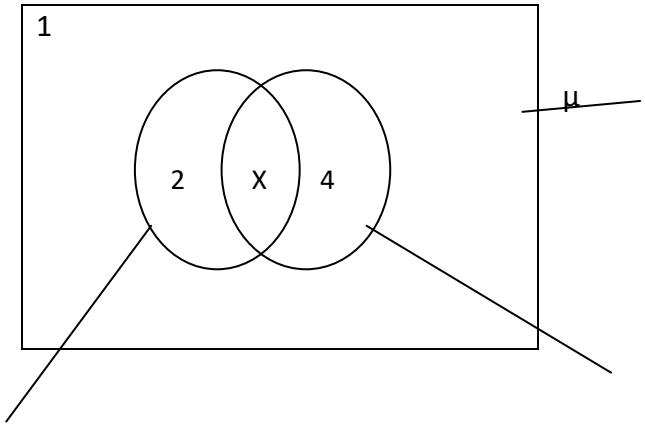
$^{6}/_{10} X\ ^{4}/_{10} +\ ^{4}/_{10} X\ ^{6}/_{10}$

$\frac{24}{100} + \frac{24}{100}$

$= \frac{48}{100}$

$= \frac{12}{25}$----------01mark

(b). By using veen diagram



---------- 01mark

μ= 20

1 + 2 + X + 4 = 20

X + 7= 20

X= 13 ----------01mark

Pupils study History 13 + 14

= 17 ---------- 01mark

4. (a). y= 3 − 4X passes through the points (-8,5)

y= -4 X + 3

y= MX + C

M_1= -4 ---------- 01mark

Perpendicular lines M_1M_2= -1

-4M_2= -1

M_2= $^1/_4$ ---------- 0.5marks

$^1/_4$= $\frac{y-5}{X-(-8)}$

4y – 20 = X + 8 ---------- 0.5mark

4y= X + 28

Y= $\frac{X}{4}$ + 7 ---------- 01marks

(b). Ç = a + b but a = ɑ + b but ɑ = 2i b= -i+ 2j

Č = oi + 2j + -j + 2j ----------01mark

= -I + 4j ---------- 01mark

|Ç| = $\sqrt{-(1)^2 + (4)^2}$ = $\sqrt{15}$ ---------- 01mark

5. (a). Given length of two sides 18cm and 24cm and included angle 30°

Area = $^1/_2$ $ab \sin \hat{C}$ ---------- 01mark

$^1/_2$ X 18 X 24 $\sin 30$ ---------- 0.5mark

9 X 24 X O.5 ---------- 0.5mark

108cm^2----------01mark

(b). Interior angle + exterior angle= 180

Let X be exterior angle

X + 60 + X = 180 ---------- 0.5mark

2X + 60 = 180

2X= 120

X= 60 ---------- 01mark

Exterior angle= 60° and interior angle = 120°

Umber of side n= $\frac{360}{exterior\ angle}$ ---------- 0.5mark

n= $\frac{360}{60}$

n= 6

The polygon has 6 sides---------- 01mark

6. (a). 18 litre = 154Km
X litre = 231Km ? ---------- 01mark

154X = 18 X 231----------01 mark

X= $\frac{4154}{154}$

X= 27 Litre

27 Litre would be used for 231Km----------01mark

(b) X α $\frac{y}{Z}$ ---------- 0.5mark

X= $\frac{Ky}{Z}$ X= 8, y= 12, Z= 6
X? y= 16, Z= 4

8= $\frac{K(12)}{6}$ ---------- 0.5mark

2K= 8 K= 4 ---------- 01mark

X= $\frac{4(16)}{4}$

X= 16

The value of X is 16----------01mark

7. (a). (i)Profit made $^{10}/_{100}$ X 5000,000

= 500,000 ---------- 01mark

(ii). The selling price = Buying price + profit made

= 5,000,000 + 5000,000 ---------- 01mark

5,500,000/= ---------- 01mark

(b). Dr Trading, Profit & loss a/c as at 31st Dec 2003 Cr

Purchase	2,200,000	Sale	2,130,000
		Gross loss c/d	70,000
	2,200,000		2,200,000
Gross loss c/f	70,000		
Expenses			
Rent	100,000		
Salary	60,000		
Wages	50,000	Net loss	280,000
	280,000		280,000

8. (a). $\log_3 x + \log_9 x + \log_{81} x + -----$

$\log_3 x + \frac{\log x}{\log_3 2} + \frac{\log x}{\log_3 4} + -----$

$\log_3 x + {}^1/_2 \log_3 x + {}^1/_4 \log_3 x + ----------$ –0.5mark

r=$\frac{{}^1/_2 \log_3 x}{\log_3 x} = \frac{{}^1/_4 \log_3 x}{{}^1/_2 \log_3 x}$ ---------- 0.5mark

r= ${}^1/_2$ n= 10 $G_1 = \log_3 x$ ---------- 0.5mark

Sn= $\frac{G_1(1-r^n)}{1-r}$ ---------- 0.5mark

$S_{10} = \log_3 x \frac{(1-({}^1/_2))}{1-{}^1/_2}$ ---------- 0.5mark

= $\frac{255}{128} \log_3 x$

The sum of the first 10 term of the series = $\frac{255}{128} \log_3 x$ ---------- 0.5mark

(b). Given n= 2 p= 5000,000

R= 10% T= 1 ---------- 0.5mark

An= $P(1 + \frac{RT}{100})^n$ ---------- 0.5mark

A_2= $5000,000(1 + \frac{10\times1}{100})^2$

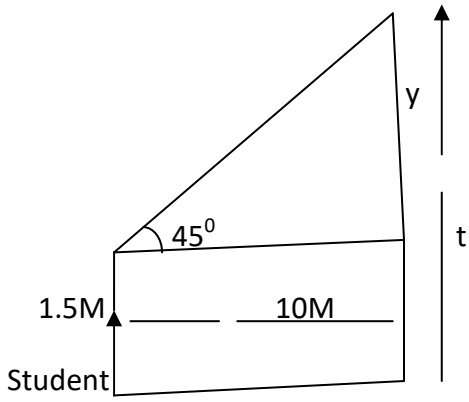
= $500000(1 + 0.1)^2$---------- 0.5mark

= $500,000(1.1)^2$

= $500,000 \times 1.21$ ---------- 0.5mark

The amount of money acculated at the end of 2 year is 605,000/= ---------- 01mark

9. (a).



---------- 01mark

Tan 45^0= $\frac{y}{10}$

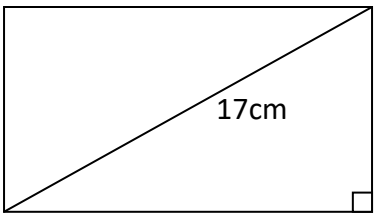
y= (Tan 45)10 ---------- 01mark

= 1 X 10

=10M

The height of the tree 1.5M + 10M= 11.5M----------01mark

(b). Rectangular frame



---------- 01mark

Length= 15cm

By using Pythagoras theorem

$15^2 + y^2 = 17^2$

$225 + y^2 = 289$ ---------- 01mark

$y^2 = 289 - 225$

$y^2 = 64$

$y = 8$

The width of the frame = 8cm ---------- 01mark

10. (a). $\frac{2X^2}{2} + \frac{X}{2} - \frac{1}{2} = 0$

$X^2 + \frac{X}{2} - \frac{1}{2} = 0$

a=1 b= $^1/_2$ b= $-^1/_2$

$X^2 + {}^1/_2 X = {}^1/_2$ ---------- 0.5mark

Add$({}^1/_2\, b)^2$ to the both side

$({}^1/_2 \times {}^1/_2)^2 = ({}^1/_4)^2$ ---------- 0.5mark

$X^2 + {}^1/_2 \times ({}^1/_4)^2 = {}^1/_2 + ({}^1/_4)^2$

$X^2 + {}^1/_2\, X + ({}^1/_4)^2 = {}^1/_2 + {}^1/_{16}$ ---------- 0.5mark

$(X + {}^1/_4)^2 = \frac{8+1}{16}$

$(X + {}^1/_4)^2 = \frac{9}{16}$

$\sqrt{(X + {}^1/_4)^2} = \pm\sqrt{{}^9/_{16}}$ ---------- 0.5mark

$X + {}^1/_4 = \pm {}^3/_4$ ---------- 0.5mark

$X = -{}^1/_4 \pm {}^3/_4$

$X = -{}^1/_4 + {}^3/_4 = {}^1/_2$ or

X= $-{}^{1}/_{4} - {}^{3}/_{4} = -1$

X= ${}^{1}/_{2}$ or -1----------0.5mark

(b). $a^2 + b^2 = (a + b)(a-b)$ ---------- 01mark

$2557^2 - 2547^2 = (2557 + 2547)(2557 - 2547)$

$= (5104)10$

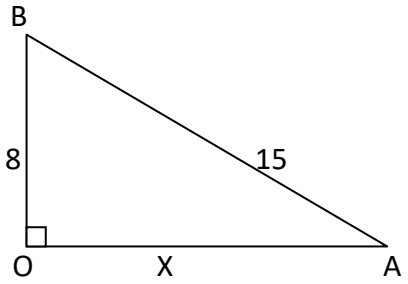
$= 51040$ ---------- 01mark

The value of $2557^2 - 2547^2$

= 51040 ---------- 01mark

SECTION B (40 MARKS)

11. (a).



---------- 0.5mark

By using Pythagoras theorem

$a^2 + b^2 = c^2$ ---------- 0.5mark

$8^2 + X^2 = 15^2$

$64 + X^2 = 225$

$X^2 = 225 - 64$ ---------- 01mark

$X^2 = 161$

$X = \sqrt{161}$

X= 12.69 unit ---------- 01mark

(b). (i) Given A= 75

Marks	f Frequency	X Class mark	x−A (d)	fd	c.f
40-50	10	45	-30	-300	10
50-60	22	55	-20	-440	32
60-70	34	65	-10	-340	66
70-80	25	75	0	0	91
80-90	7	85	10	70	98
90-100	2	95	20	40	100
	100			$\varepsilon fd = -970$	

----02mark

$\overline{X} = A + \frac{\varepsilon fd}{\varepsilon f}$ ---------- 0.5mark

$= 75 + \frac{-970}{100}$ ---------- 0.5mark

$= 75 - 9.7$

$\overline{X} = 65.3$ ---------- 01mark

(ii). Median = $L + \frac{(\frac{N}{2} - Nb)i}{Nw}$ ---------- 0.5mark

Median class = 60 – 70

i= 60, N= 100 Nb, 32

Nw= 34 i= 10

Median = $60 + \frac{(\frac{100}{2} - 32)10}{34}$ ---------- 0.5mark

$= 60 + (\frac{50 - 32}{34})10$

$= 60 + (\frac{18}{34})10$ ---------- 01mark

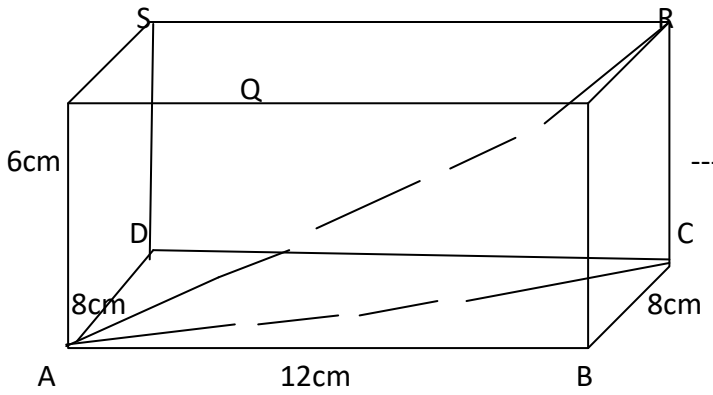
$60 + \frac{180}{34}$

60 + 5.29

Median = 65.29 ---------- 01mark

12. (a). (i). Cuboid ABCDPQRS

P



------- 01mark

$\overline{AC}^2 = \overline{AB}^2 + \overline{BC}^2$ ---------- 01mark

$\overline{AB} = 12^2 + 8^2$

$= 144 + 64$

$=208$ ---------- 01mark

$\overline{AB} = \sqrt{208} = 14.42 = 14cm$

ABCD

$$Tan\ A = \frac{oppositr}{adjuacent}$$

$= \frac{6}{\sqrt{208}}$ ----------0.5mark

Tan A= 0.4161

$A = \tan^{-1}(0.4161)$

A=23° ----------0.5mark

(b). Distance along the parallel latitude

$d = \frac{\pi RQCOS\alpha}{180}$ KM ---------- 0.5mark

Given d= 1500Km

$\alpha = 60N$

Q=? ---------- 0.5mark

1500= $\frac{3.14\ X\ 6370\ X\ QCOS60}{180}$ ---------- 01mark

270000= 3.14 x 6370 x Q x 0.5

270000= 10000.9Q ---------- 01mark

Q=$\frac{270000}{10000.9}$

Q= 26.99

Q= 27°

Difference between their longitude=27° ---------- 01mark

13. (a). B=$\begin{pmatrix} x-2 & x-2 \\ 3 & x \end{pmatrix}$

Matrix B has no inverse determinant=0 ---------- 0.5mark

$|B| = (x-2)x - 3(x-2)$

0= $x^2 - 2x - 3x + 6$

$x^2 - 5x + 6 = 0$ ---------- 01mark

$x^2 - 3x - 2x + 6 = 0$

$x(x-3) - 2(x-3) = 0$ ---------- 0.5mark

$(x-2)(x-3) = 0$

$x - 2 = 0\ or\ x - 3 = 0$

$x = 2\ or\ x = 3$ ---------- 01mark

(b). $5x - 3x = 7$
$2x - y = 1$

$\begin{pmatrix} 5 & -3 \\ 2 & -1 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 7 \\ 1 \end{pmatrix}$

Let A= $\begin{pmatrix} 5 & -3 \\ 2 & -1 \end{pmatrix}$

$|A| = 5x - 1 - 2x - 3$

= -5 – (-6)

= -5 + 6 ---------- 01mark

= 1

$|A| = 1$

$A^{-1}=\frac{1}{1}\begin{pmatrix}-1 & 3\\ -2 & 5\end{pmatrix}$---------- 01mark

$A^{-1}=\begin{pmatrix}-1 & 3\\ -2 & 5\end{pmatrix}$

Pre- multiply byA^{-1} to the both side

$\begin{pmatrix}-1 & 3\\ -2 & 5\end{pmatrix}\begin{pmatrix}5 & -3\\ 2 & -1\end{pmatrix}\begin{pmatrix}x\\ y\end{pmatrix}=\begin{pmatrix}-1 & 3\\ -2 & 5\end{pmatrix}\begin{pmatrix}7\\ 1\end{pmatrix}$ ---------- 01mark

but $A^{-1}A$= Identify matrix

$\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}\begin{pmatrix}x\\ y\end{pmatrix}=\begin{pmatrix}-1\times 7+3\times 1\\ -2\times 7+5\times 1\end{pmatrix}$

$\begin{pmatrix}x\\ y\end{pmatrix}=\begin{pmatrix}-4\\ -9\end{pmatrix}$

$x = -4\ and\ y = -9$ ---------- 01mark

(c). Given the points $(2,3)\ into\ (8,18)$

(-3,-2) into (-7,-17)

Let matrix T= $\begin{pmatrix}a & b\\ c & d\end{pmatrix}$

$\begin{pmatrix}x^1\\ y^1\end{pmatrix}=\begin{pmatrix}a & b\\ c & d\end{pmatrix}\begin{pmatrix}x\\ y\end{pmatrix}$ ---------- 0.5mark

$\begin{pmatrix}8\\ 18\end{pmatrix}=\begin{pmatrix}a & b\\ c & d\end{pmatrix}\begin{pmatrix}2\\ 3\end{pmatrix}\ and\ \begin{pmatrix}-7\\ -17\end{pmatrix}=\begin{pmatrix}a & b\\ c & d\end{pmatrix}\begin{pmatrix}-3\\ -2\end{pmatrix}$

$8 = 2a\ ) + 3b \ldots\ldots\ldots\ (i) -7 = -3a - 2b \ldots\ldots\ldots (iii)$
$18 = 2c + 3d \ldots\ldots\ldots (ii)\ \ -17 = -3c - 2d \ldots\ldots (iv)$

$2\left|\begin{array}{l} 2a + 3b = 8 \ldots\ldots\ldots (i)\\ 3a - 2b = -7 \ldots\ldots\ldots (iii)\end{array}\right.\quad 3\left|\begin{array}{l} 2c + 3d = 18 \ldots\ldots\ldots (ii)\\ 2c - 2d = -17 \ldots\ldots\ldots (iv)\end{array}\right.$

$+\left|\begin{array}{l} 6a + 9b = 24\\ -6a - 4b = -14\end{array}\right.\qquad +\left|\begin{array}{l} 6c + 9d = 54\\ -6c - 4d = -34\end{array}\right.$

$5b = 10$ $\ldots\ldots\ldots.0.5mark$
$b = 2$

$5d = 20$ $\ldots\ldots\ldots 0.5mark$
$d = 4$

$2a + 3(2) = 8$
$2a + 6 = 8$ $\ldots\ldots\ldots\ldots 0.5mark$
$2a = 2$

$a = 1$

$2c + 3(4) = 18$
$2c + 12 = 18$
$2c = 6$

$c = 3$

The matrix T= $\begin{matrix} 1 & 2 \\ 3 & 4 \end{matrix}$ ---------- 01mark

14. Given f(x)= $\begin{cases} -2 \; if \; x < 0 \\ 4 \; if \; x = 0 \\ x + 1 \; if \; x > 0 \end{cases}$

Table of values

$y = -2, x < 0$

X	0	-1	-2	-3	-4
y	-2	-2	-2	-2	-2

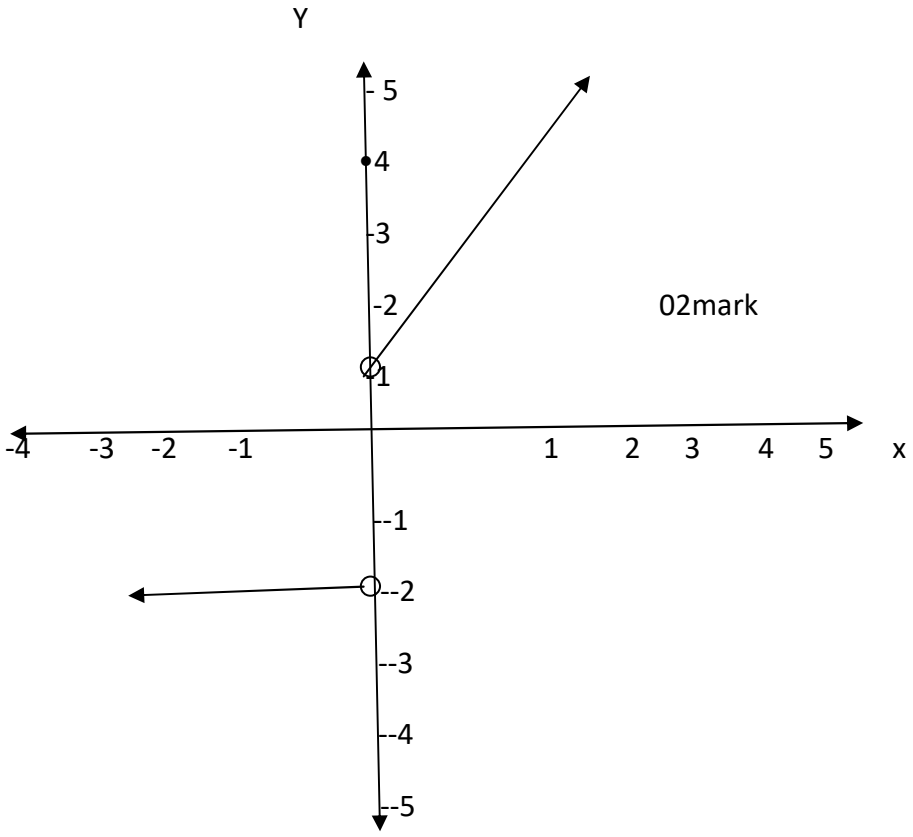
0.5mark

$y = 4 \quad x = 0$
$(0,4)$

$y = x + 1 \; if \; x > 0$

X	0	1	2	3	4	5
y	1	2	3	4	5	6

0.5mark



$Domain = \{all\ real\ number\ x\} \ldots\ldots\ldots\ldots\ldots 0.5mark$

$Domain = \{y: y \geq 1\ and\ y = -2\} \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots .0.5mark$

(b). Linear inequalities

$let\ x\ be\ tins\ of\ tomato$

$y\ be\ tins\ of\ chilli\ source$

$960x + 768y \leq 4800 \ldots\ldots\ldots . (i)$

$x + y \leq 6 \ldots\ldots\ldots\ldots\ldots (ii)$ ---------- 0.5mark

$x \geq 1 \ldots\ldots\ldots\ldots\ldots\ldots . . (iii)$

$y \geq 1 \ldots\ldots\ldots\ldots . . (iv)$

Objective function f(x,y)=384 + 192y

Table of value

$960x + 768y = 4800$

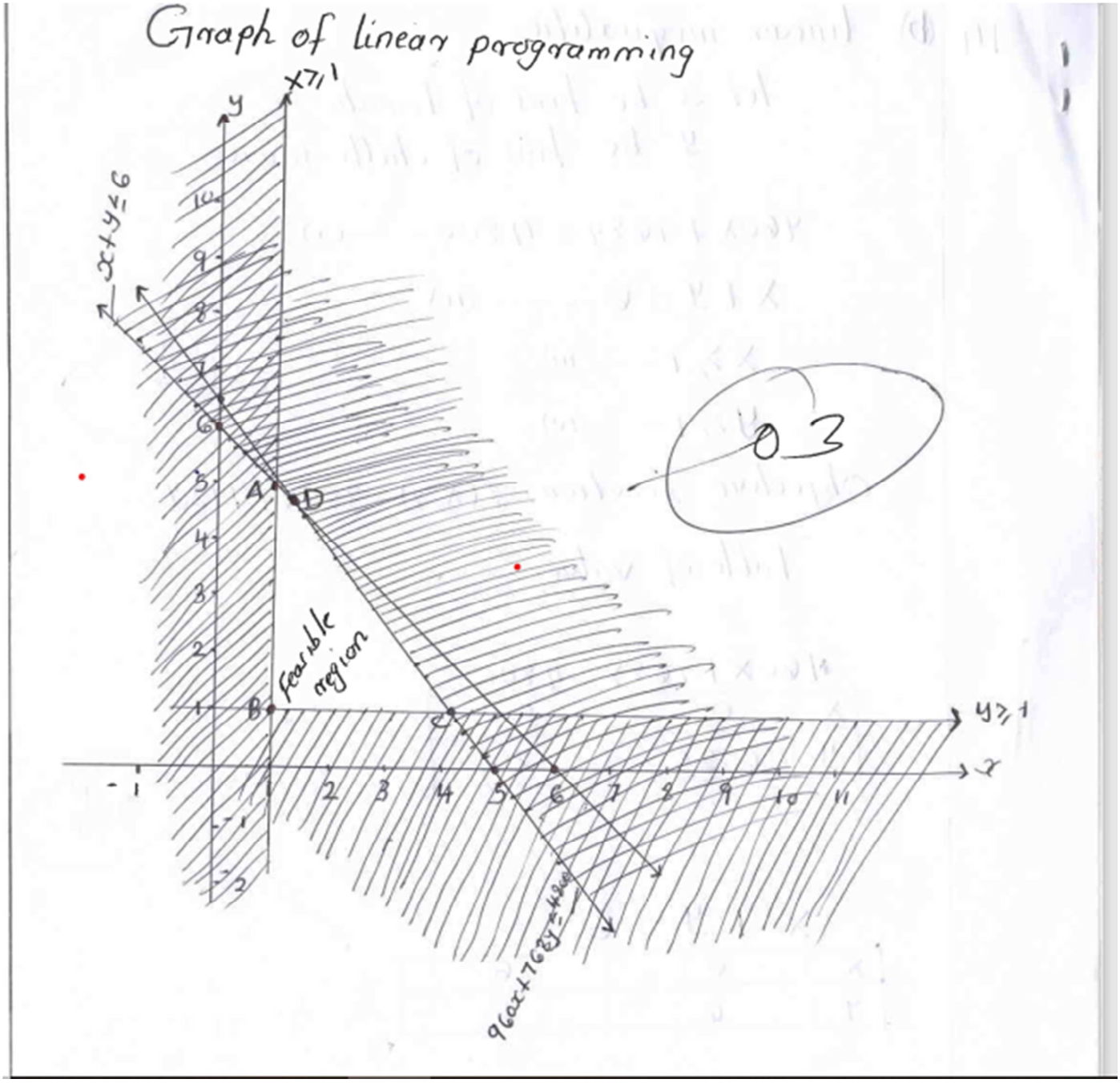
X	0	5
Y	6.25	0

$$x + y = 6$$

X	0	6
Y	6	0

All table 0.5mark

Graph of linear programming



Corner points

Points	F(x,y)=384x + 192y	value
A(1,5)	383(1) + 192(5)	1344
B(1,1)	384(1) + 192(1)	576
C(4.2,1)	384(4.2) + 192(1)	1804.8
D(1,5)	384(1) + 192(5)	576

01mark

The wholesaler must sell 4.2 dozens of tomato and 1 dozen of chilli source to get maximum profit of 1804.8/= 01mark